

Week 6 – Data Visualization Mastery with Seaborn

Interactive Sales Dashboard

1. Project Overview

This project was completed as part of Week 6 of my Data Science Internship at The Developers Arena.

The objective is to analyze sales data and create meaningful statistical visualizations using Seaborn. An interactive sales dashboard is also created using Plotly to present sales trends, product performance, and regional performance.

2. Project Objectives

- Analyze the provided sales dataset.
- Perform data cleaning and preparation.
- Create statistical visualizations using Seaborn.
- Create at least five different types of charts.
- Analyze product and regional sales.
- Analyze sales trends over time.
- Create an interactive dashboard using Plotly.
- Present important findings through visualizations.
- Document the complete project workflow.

3. Dataset

The project uses:

sales_data.csv

The dataset contains 100 rows and 7 columns.

Columns:

- Date
- Product
- Quantity
- Price
- Customer_ID
- Region
- Total_Sales

4. Dataset Validation

The dataset was checked before analysis.

Results:

- Rows: 100
- Columns: 7
- Missing values: 0
- Duplicate rows: 0
- Products: 5
- Regions: 4

The Date column was converted to datetime format for time-based analysis.

5. Technologies Used

- Python
- Pandas
- Seaborn
- Matplotlib
- Plotly
- Jupyter Notebook
- Visual Studio Code

6. Project Structure

Task_6_Data_Visualization_Mastery_with_Seaborn/

```
|
|— dashboard.ipynb
|— dashboard.py
|— sales_data.csv
|— requirements.txt
|— dashboard_demo.gif
|
|— data/
|
|— visualizations/
|   |— sales_by_product.png
|   |— sales_trend.png
|   |— sales_by_region.png
|   |— sales_distribution.png
|   |— correlation_heatmap.png
```

```
|  
|— screenshots/  
|  
|— report/  
|   — analysis_report.md
```

7. Setup Instructions

Install the required Python libraries:

```
python -m pip install -r requirements.txt
```

Run the dashboard using:

```
python dashboard.py
```

The Jupyter Notebook can also be opened using:

```
jupyter notebook dashboard.ipynb
```

8. Data Analysis

Pandas was used for:

- Loading the CSV file.
- Checking dataset structure.
- Checking missing values.
- Checking duplicate records.
- Converting dates.
- Grouping sales data.
- Calculating total sales.
- Calculating quantities.
- Preparing data for visualization.

Basic results from the dataset:

- Total Sales: 12,365,048
- Total Quantity Sold: 478
- Number of Products: 5
- Number of Regions: 4

9. Sales by Product

Sales were grouped by product to compare product performance.

The analysis shows that Laptop has the highest total sales among the available products.

A Seaborn bar chart was created to make the product comparison easier to understand.

10. Sales by Region

Sales were grouped by region to compare regional performance.

The North region generated the highest total sales among the four regions.

A regional sales chart was created using Seaborn.

11. Monthly Sales Trend

The Date column was converted into datetime format and used for time-based analysis.

Monthly sales were calculated and visualized using a line chart.

The chart helps identify changes in sales performance over time.

12. Sales Distribution

Distribution charts were created to understand the spread of sales values.

Histogram, box plot, or violin plot visualizations were used where appropriate.

These visualizations help identify the distribution and variation of sales values.

13. Correlation Analysis

A correlation heatmap was created using Seaborn.

The heatmap helps understand relationships between numerical columns such as:

- Quantity
- Price
- Total_Sales

This provides a quick overview of relationships between numerical variables.

14. Seaborn Visualizations

The project includes at least five meaningful visualizations:

1. Sales by Product – Bar Chart
2. Monthly Sales Trend – Line Chart
3. Sales by Region – Bar Chart
4. Sales Distribution – Statistical Chart
5. Correlation Heatmap

These charts provide different views of the sales dataset.

15. Interactive Dashboard

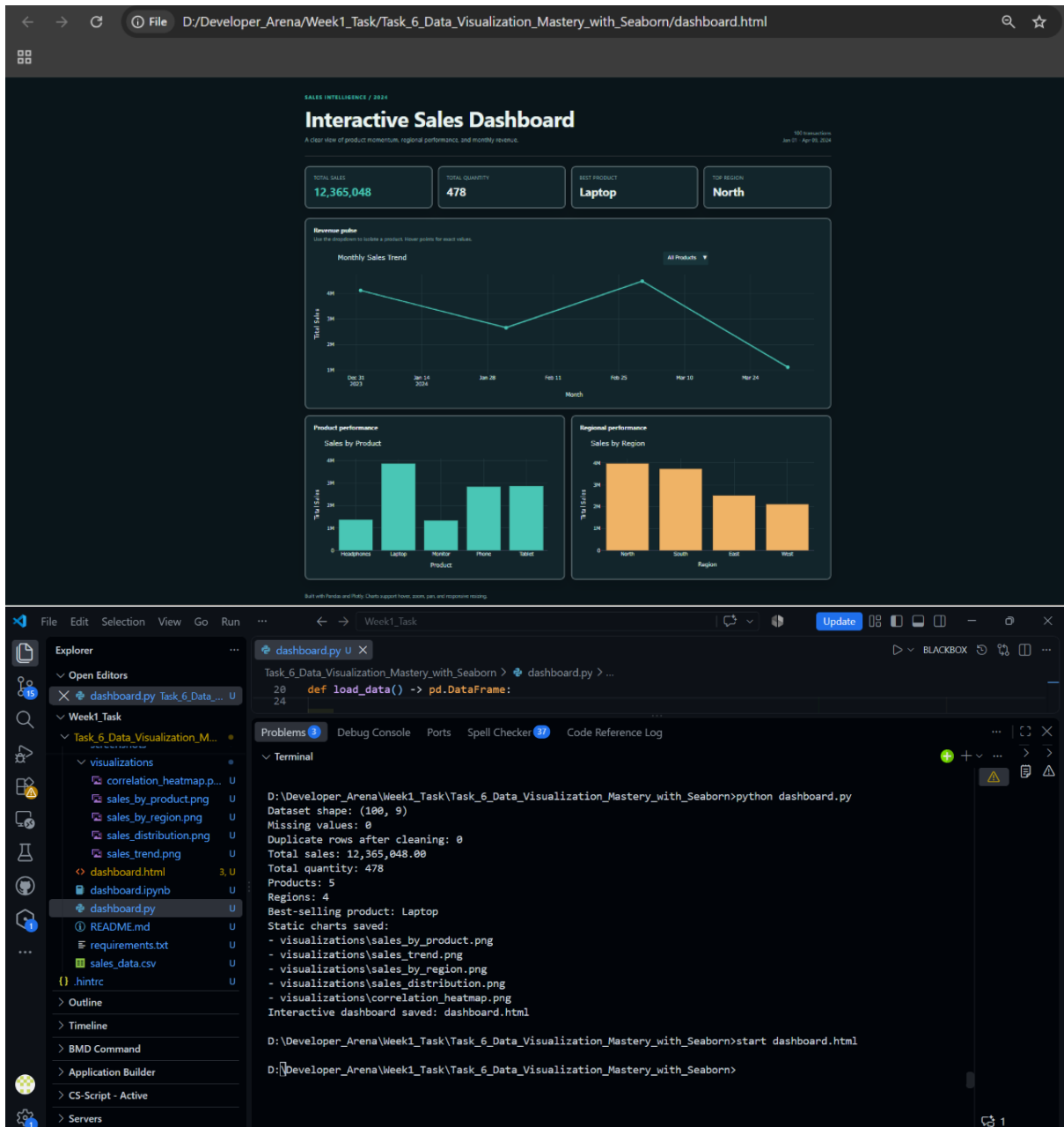
An interactive dashboard was created using Plotly.

The dashboard includes:

- Total Sales KPI
- Total Quantity KPI

- Number of Products
- Number of Regions
- Product sales visualization
- Regional sales visualization
- Monthly sales trend
- Interactive chart elements

Plotly provides interactive features such as hover information, zooming, and filtering where applicable.



16. Key Insights

- The dataset contains 100 sales transactions.
- Laptop is the highest-performing product by total sales.

- North is the highest-performing region.
- Total sales in the dataset are 12,365,048.
- A total quantity of 478 units was recorded.
- Sales performance changes across the analyzed dates.
- The visualizations make product, regional, and sales trends easier to compare.

17. Error Handling and Validation

The project includes basic validation for:

- Missing files
- Missing values
- Duplicate records
- Invalid data
- Date conversion
- Required columns

The dataset was validated before performing the analysis.

18. Testing

The project was tested by running:

`python dashboard.py`

The following were verified:

- Dataset loads correctly.
- Data processing works correctly.
- Required visualizations are generated.
- Dashboard loads successfully.
- Interactive dashboard elements work correctly.
- No major runtime errors remain.

19. Learning Outcomes

Through this project, I learned:

- Seaborn data visualization.
- Statistical charts.
- Customizing charts and labels.
- Correlation heatmaps.
- Distribution visualizations.
- Creating multiple visualizations.

- Using Plotly for interactive charts.
- Designing a dashboard layout.
- Combining data analysis and visualization.

20. Conclusion

This project demonstrates how Python, Pandas, Seaborn, and Plotly can be used together to analyze and visualize sales data.

The project successfully presents product performance, regional sales, sales trends, statistical distributions, and numerical relationships through multiple visualizations and an interactive dashboard.

The dashboard provides a clear and interactive way to understand the main patterns in the sales dataset.